

Assignment - 2

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$$1) i) \int_0^{2a} \int_0^{\sqrt{2a^2-n^2}} x^2 dy dn$$

$$ii) \int_0^{2a} \left[x^2 y \right]_{y=0}^{y=\sqrt{2a^2-n^2}} dn$$

$$\int_0^{2a} \left[x^2 (\sqrt{2a^2-n^2}) \right] dn$$

$$\int_0^{2a} \left[(\sqrt{2a} \sin \theta)^2 \sqrt{2a(1-\sin^2 \theta)} - (\sqrt{2a} \sin \theta)^2 \cos \theta \right] d\theta$$

$$\int_0^{\pi/2} \left[2a \sin^2 \theta \cdot \sqrt{2a-2a \sin^2 \theta} \times \sqrt{2a} \cos \theta \right] d\theta$$

$$\Rightarrow 2a \int_0^{\pi/2} \frac{1}{2} (1-\cos 2\theta) \sqrt{2a-2a \sin^2 \theta} \cdot \cos \theta d\theta$$

$$\Rightarrow a \int_0^{\pi/2} \cos \theta - \cos 2\theta \cos \theta \sqrt{2a-a \sin^2 \theta} d\theta$$

$$\cos 2\theta \cdot \cos \theta = \frac{1}{2} (2 \cos^2 \theta - 1) \cos \theta$$

$$= \frac{1}{2} (2 \cos^3 \theta - \cos \theta)$$

$$= \cos^3 \theta - \frac{1}{2} \cos \theta$$

$$\begin{aligned} x &= \sqrt{2a} \sin \theta \\ dn &= \sqrt{2a} \cos \theta d\theta \end{aligned}$$

$$\sqrt{2a} \sin \theta = 0$$

$$\sin \theta = 0$$

$$\theta = 0 \Rightarrow 0 = 0$$

$$\theta = \frac{\pi}{2}$$

$$\begin{aligned} \sqrt{2a} \sin \theta &= 2a \\ \sin \theta &= \frac{2a}{\sqrt{2a}} \end{aligned}$$

$$\theta = \frac{\pi}{2}$$

$$ii) \int_0^a \int_y^a \frac{xdxdy}{\sqrt{x^2+y^2}}$$

$$a \quad a$$

$$\frac{1}{2} \int_0^a \int_y^a \frac{x}{\sqrt{u}} \frac{du}{x} dy$$

$$\frac{1}{2} \int_0^a \int_{u=2y^2}^{u=a^2+y^2} (u)^{-1/2} du dy$$

$$\Rightarrow \frac{1}{2} \int_0^a 2 \left[u^{1/2} \right]_{u=2y^2}^{u=a^2+y^2} dy$$

$$\Rightarrow \frac{1}{2} \int_0^a 2 \left[\sqrt{a^2+y^2} - \sqrt{2y^2} \right] dy$$

$$\Rightarrow \int_0^a \left[\sqrt{a^2+y^2} - \sqrt{2} y \right] dy$$

$$\Rightarrow \left[\frac{y}{2} \sqrt{y^2+a^2} + \frac{a^2}{2} \log |y + \sqrt{y^2+a^2}| - \frac{\sqrt{2} y^2}{2} \right]_0^a$$

$$\left[\frac{a}{2} \sqrt{(2a^2)} + \frac{a^2}{2} \log |a + \sqrt{2a^2}| - \frac{a^2}{2} \log(\sqrt{a^2}) \right]$$

 $\frac{du}{dx} = 2x$

$$u = x^2 + y^2$$

$$du = 2x dx$$

$$dx = \frac{du}{2x}$$

$$u = 0 \text{ to } 20$$

$$u = y^2$$

$$u = x^2 + y^2$$

$$x = y$$

$$y = x^2 + y^2$$

$$u = y^2 + y^2$$

$$u = 2y^2$$

$$x = a$$

$$u = a^2 + y^2$$

$$\frac{a}{2} \sqrt{a^2+a^2} + \frac{a^2}{2} \log |a + \sqrt{2a^2}|$$

$$\log |a + \sqrt{2a^2}| + C$$

$$\frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$x^2 + y^2 = \frac{x^2 + y^2}{2}$$

$$y = a \quad y = 0$$

$$-\left[\frac{y^2}{\sqrt{2}} \right]_0^a$$

$$\rightarrow \left[\frac{\sqrt{2}a^2}{2} + \frac{a^2}{2} \log(a + \sqrt{2}a) \right] - \frac{a^2}{2} \log(a) - \left[\frac{a^2}{\sqrt{2}} \right]$$

$$\Rightarrow \cancel{\frac{a^2}{\sqrt{2}}} + \frac{a^2}{2} \log(a(1+\sqrt{2})) - \frac{a^2}{2} \log a - \cancel{\frac{a^2}{\sqrt{2}}}$$

$$\Rightarrow \frac{a^2}{2} \left[\log a(1+\sqrt{2}) - \log a \right]$$

$$\Rightarrow \frac{a^2}{2} \log \left| \frac{a(1+\sqrt{2})}{a} \right|$$

$$\Rightarrow \frac{a^2}{2} \log(1+\sqrt{2}) //$$

2) $\int_0^2 \int_0^y \int_{z=n-y}^{z=n+y} (n+y+z) \, dz \, dy \, dx$

$\int_0^2 \int_0^y \left[\frac{z^2}{2} + nz + xz \right]_{z=n-y}^{z=n+y} dz \, dy \, dx$

$\int_0^2 \int_0^y \left[\frac{z^2}{2} + nz + x(n+y) - \left(\frac{z^2}{2} + nz + x(n-y) \right) \right] dy \, dx$

$\int_0^2 \int_0^y xy \, dy \, dx \rightarrow$

$$\Rightarrow 2 \int_0^2 \left[\frac{x^2}{2} \right]_0^y y dy$$

$$\Rightarrow \frac{2}{2} \int_0^2 y^3 dy \Rightarrow \left[\frac{y^4}{4} \right]_0^2$$

$$\Rightarrow \frac{1}{4} (16) \Rightarrow \underline{\underline{4}} \Rightarrow 16$$

ii) $\int_0^{\log 2} \int_0^n \int_0^{n+\log y} e^{x+y+z} dz dy dx$

$$= \int_0^{\log 2} \int_0^n e^{x+y+z} \left[xz + yz + \frac{z^2}{2} \right]_{z=0}^{z=n+\log y} dy dx$$

$$\Rightarrow \int_0^{\log 2} \int_0^n e^{x+y+z} \left[x(n+\log y) + y(n+\log y) + \frac{n^2 + 2n \log y + \log^2 y}{2} \right] dy dx$$

$$\Rightarrow \int_0^{\log 2} \int_0^n e^{x+y+z} \cdot \left(\frac{n^2 + 2n \log y + \log^2 y}{2} + ny + y \log y \right) dy dx$$

$$\Rightarrow \int_0^{\log 2} \left[(e^{2x} (x^2 - \frac{1}{2}) - e^x \cdot e^x) + e^{2x} + e^x \right]$$

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$$\int_0^{8\log 2} \left[\cancel{e^{2n}} (\cancel{2n} - \cancel{1}) \frac{2e^{3n}}{3} - \frac{4e^{3n}}{9} + e^n \right]_0^{8\log 2}$$

$$\left(\frac{8\log 2}{3} - \frac{4e}{9} + 2 \right) + \frac{4}{9}$$

$$\frac{8\log 2}{3} - \frac{28}{9} + 1 = \frac{8\log 2}{3} - \frac{10}{9}$$

$$= \left[\frac{2t^{3/2}}{3} \right]$$

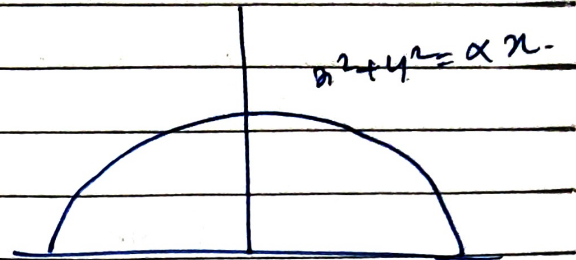
3) Evaluate: $\int \int (\alpha^2 - x^2 - y^2)^{1/2} dx dy$ region α

is bounded by the semicircle $x^2 + y^2 = \alpha x$ in the first octant

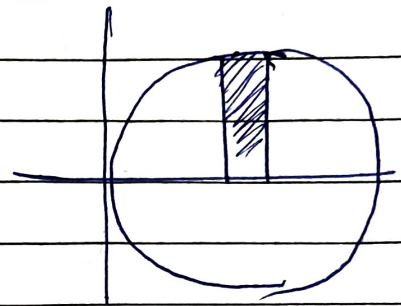
$$x^2 + y^2 = \alpha x$$

$$x^2 - \alpha x + \alpha^2 - \alpha^2 + y^2 = 0$$

$$(x - \alpha/2)^2 + y^2 = \alpha^2/4$$



$$\int_0^{\alpha} \int_0^{\sqrt{\alpha^2 - x^2 - y^2}} \sqrt{\alpha^2 - x^2 - y^2} dy dx$$



$$\left[\frac{y}{2} \sqrt{\alpha^2 - x^2 - y^2} + \frac{\alpha^2}{2} \sin^{-1} \frac{y}{\alpha} \right]_0^{\sqrt{\alpha^2 - x^2 - y^2}}$$

$$\Rightarrow \int_0^{\pi/2} \int_0^{\alpha \cos \theta} r \sqrt{\alpha^2 - r^2} dr d\theta$$

$$t = \alpha^2 - r^2$$

$$dt = -2r dr$$

$$\frac{dt}{-2} = r dr$$

$$t \cos \theta = \alpha^2 \sin \theta$$

$$t = \alpha^2$$

$$-\frac{1}{2} \int_0^{\pi/2} \int_{\alpha^2}^{\alpha^2 \sin^2 \theta} r^2 (-r^2 \sqrt{t} dt d\theta) = \left[\frac{2t^3}{3} \right]_{\alpha^2}^{\alpha^2 \sin^2 \theta}$$

$$t \cos \theta = \alpha^2 \sin \theta$$

$$t = \alpha^2$$

$$\left[\frac{2t^{3/2}}{3} \right]_{\alpha^2}^{\alpha^2 \sin^2 \theta} = \frac{1}{2} \int_0^{\pi/2} \left[\frac{2 \alpha^3 \sin^3 \theta}{3} - \frac{\alpha^3}{2} \right] d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \left[-\alpha^3 \sin^3 \theta + \alpha^3 \right] d\theta \Rightarrow \frac{\alpha^3}{3} \left[\frac{\pi}{2} - \frac{2}{3} \right]$$

6) Find Volume bounded above by plane $z=2$ and sphere $x^2+y^2+z^2=9$.

sol

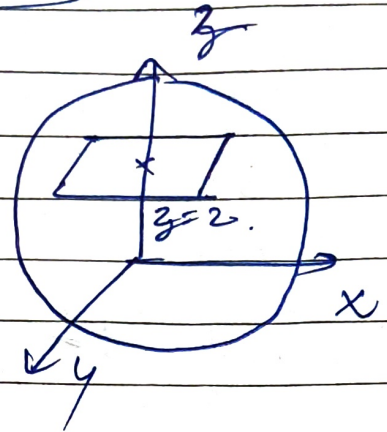
$$x^2+y^2+z^2=9 \quad \text{--- (1)} \quad \boxed{r=3}$$

$$z^2 = 9 - x^2 - y^2$$

$$z = \sqrt{9 - x^2 - y^2}$$

Units :

$r \geq 0 \rightarrow r = 3$
$\theta \geq 0 \rightarrow \theta = 2\pi$
$z = 2 \rightarrow z = \sqrt{9 - x^2 - y^2}$



$$d\text{volume} = r dr d\theta dz$$

$$\int_{\theta=0}^{2\pi} \int_{r=0}^3 \int_{z=2}^{\sqrt{9-x^2-y^2}} dz r dr d\theta$$

Let

$$A = \sqrt{9 - x^2 - y^2}$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\theta=0 \quad r=0$$

$$\int_{\theta=0}^{2\pi} \int_{r=0}^3 (A - 2) r dr d\theta \rightarrow \int_{\theta=0}^{2\pi} \int_{r=0}^3 (Ar - 2r) dr d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[\frac{A}{2} [r^2]_0^3 - 2 \left[\frac{r^2}{2} \right]_0^3 \right] d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(\frac{9A}{2} - 9 \right) d\theta \rightarrow \left[\frac{9A}{2} - 9 \right] \left[\theta \right]_0^{2\pi}$$

$$= \left[\frac{9A}{2} - 9 \right] [2\pi]$$

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$$\int_{\theta=0}^{2\pi} \int_{r=0}^3 (\sqrt{9-r^2} - 2) dy dr.$$

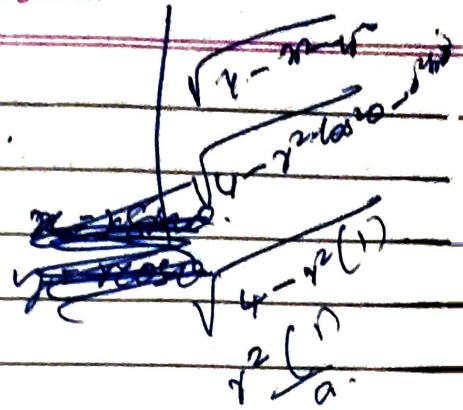
$$\int_0^{\pi/4} \int_0^2 (\sqrt{3^2 - r^2} - 2) r$$

$$4 \int_0^{\pi/2} \left[\frac{1}{12} \left(\int_0^2 \sqrt{t} - \int_0^2 2r \right) \right]$$

$$\Rightarrow 4 \int_0^{\pi/2} \left[\frac{2t^{3/2}}{3} \right]_0^2 - \left[r^2 \right]_0^2$$

$$\Rightarrow 4 \int_0^{\pi/2} \frac{(9)^{3/2}}{3} - \frac{(5)}{3} - 4$$

$$\Rightarrow \boxed{\frac{54 - 10\sqrt{5}}{3}}$$



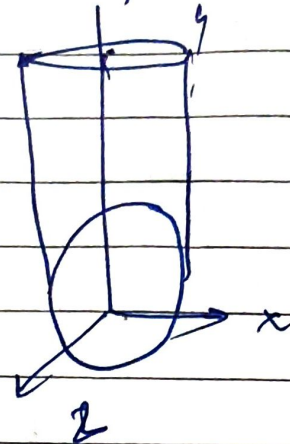
- 5) Find the volume of ^{sphere} ~~$z = x^2 + y^2$~~ $x^2 + y^2 + z^2 = 4$ and below by paraboloid $z = x^2 + y^2$

$$x^2 + y^2 + z^2 = 4 \quad (1)$$

$$\boxed{r = 2}$$

$$z^2 = 4 - x^2 - y^2$$

$$z = \sqrt{4 - x^2 - y^2}$$



$$az = x^2 + y^2 \quad (2)$$

$$z = \frac{x^2}{a} + \frac{y^2}{a}$$

$$\frac{x^2 + y^2}{a} \leq z \leq \sqrt{4 - x^2 - y^2}$$

$$\int \int \int_{z = \frac{x^2 + y^2}{a}}^{z = \sqrt{4 - x^2 - y^2}} dz dy dx$$

$$\int \int \left[z \right]_{z = \frac{x^2 + y^2}{a}}^{z = \sqrt{4 - x^2 - y^2}} dy dx$$

$$\Rightarrow \int_{\theta=0}^{2\pi} \int_{r=0}^{r=2} \left(\sqrt{4 - r^2} - \left(\frac{r^2}{a} \right) \right) \frac{dy dx}{r dr d\theta}$$

$$\Rightarrow \int_{\theta=0}^{2\pi} \int_{r=0}^{r=2} \left[\sqrt{4 - r^2} - \frac{r^2}{a} \right] r dr d\theta$$

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$$\begin{aligned} x &= r \cos \theta & y &= r \sin \theta \\ r \cos \theta &= 0 & r \sin \theta &= 0 \\ \boxed{r = \frac{\pi}{2}} & & \boxed{r = 0} & \\ dy dx &= r dr d\theta & & \end{aligned}$$

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$$\Rightarrow \int_{\theta=0}^{2\pi} \int_{r=0}^{\pi/2} \left[r\sqrt{4-r^2} - \frac{r^3}{a} \right] dr d\theta$$

Let, $4-r^2 = t$
 $dt = -2r dr$
 $dr =$

$$r\sqrt{4-r^2} - \frac{r^3}{a}$$

$$t = 4 - r^2$$

$$\frac{dt}{-2} = r$$

$$-\frac{1}{2} \int \sqrt{t} - \int \frac{r^3}{a}$$

$$= -\frac{1}{2} \int \sqrt{t} - \int_0^2 \frac{r^3}{a}$$

$$= -\frac{1}{2} \left[\frac{t^{1/2+1}}{1/2+1} \right] - \frac{1}{a} \left[\frac{r^4}{4} \right]_0^2$$

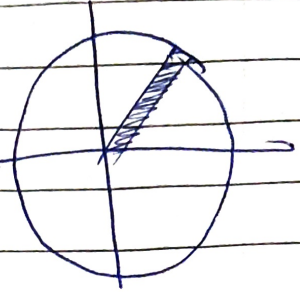
$$= -\frac{1}{2} \left[\frac{t^{3/2}}{3/2} \right] - \frac{1}{4a} [16 - 0]$$

$$= -\frac{1}{2} - t^{3/2} - \frac{4}{a}$$

$$= -t\sqrt{t} - \frac{4}{a}$$

$$4) \iint_D xy (x^2 + y^2)^{3/2} dx dy \quad x^2 + y^2 = 1.$$

$$\int_0^{2\pi} \int_0^{\pi/2} r^2 \sin \theta \cos \theta (r^2)^{3/2} r dr d\theta$$



$$\int_0^{2\pi} \left[\frac{r^7}{7} \right]_0^{\pi/2} \sin \theta \cos \theta d\theta$$

$$\frac{1}{7} \int_0^{\pi/2} \sin \theta \cos \theta d\theta$$

$$\frac{1}{7} \left[\frac{t^2}{2} \right]_0^1 = \frac{1}{14}$$

$$7) \int_0^a \int_0^{\sqrt{a^2 - y^2}} y^2 \sqrt{x^2 + y^2} dx dy$$

$$= \int_0^a \int_0^{\pi/2} r^3 \cos^2 \theta \cdot r dr d\theta$$

$$= \int_0^a \int_0^{\pi/2} r^4 \cos^2 \theta d\theta dr$$

$$= \frac{a^5}{5} \int_0^{\pi/2} \cos^2 \theta \, d\theta$$

$$= \frac{a^5}{5} \times \frac{\pi}{4} = \frac{a^5 \pi}{20}$$

8) $\int_0^{\pi/2} \int_0^{2a \cos \theta} r^2 \sin \theta \, dr \, d\theta$

$$t = \sin \theta$$

$$dt = \cos \theta \, d\theta$$

sol $\int_0^{\pi/2} \int_0^{2a \cos \theta} r^2 \sin \theta \, dr \, d\theta$

$$(2a)^3 \int_0^{\pi/2} t^3 \, dt \, d\theta$$

$$= \frac{1}{4} (2^3) (a^3) \times \pi$$

$$= \frac{1}{4} [8] a^3$$

$$= 2a^3$$

9) $x^2 + y^2 = 4z$ and plane $z = 4$.

$$= 4 \int_0^{\pi/2} \int_0^4 \int_{\frac{y^2}{4}}^4$$

$$x^2 + y^2 = 4z \text{ at } z = 4.$$

$$x^2 + y^2 = 4z$$

$$r^2 = 4z$$

$$z = \frac{r^2}{4}$$

$$z = 4$$

$$= 4 \int_0^{\pi/2} \int_0^4 \left(16 - \frac{r^4}{16} \right) dr d\theta$$

$$= 4 \int_0^{\pi/2} \left[2r^2 - \frac{r^5}{80} \right]_0^4 d\theta$$

$$= 4 \int_0^{\pi/2} 32 d\theta \Rightarrow 4 \times 32 \left[\theta \right]_0^{\pi/2} = 32\pi$$

10) $\iiint x^2 y z \, dx \, dy \, dz$

$$x^2 + y^2 = a, \quad z = 0, \quad z = 2$$

$$\iiint_0^2 x^2 y^2 z^2 dz dy dx$$

$$\iiint_0^2 \left[\frac{z^3}{3} \right]_0^2$$

$$\int_0^{\pi/2} \int_0^3 r^4 \cos^2 \theta \sin \theta d\theta$$

$$\int_0^{\pi/2} \int_0^3 \left[\frac{r^5}{5} \right]_0^3 \cos^2 \theta \sin \theta d\theta$$

$$= \frac{2 \times 243}{5} \int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta$$

$$= \frac{-2 \times 243}{5} \int_1^0 \left[\frac{t^3}{3} \right]$$

$$= \frac{-2 \times 243}{5} \times \frac{-1}{3} = \frac{168}{5}$$

$\frac{168}{5} \times 4 = \frac{672}{5}$ (For considering all coordinates)